

# E-Commerce Web Application - Project Documentation

## Professional GitHub Portfolio Version

This document provides a concise, professional overview of an E-Commerce Web Application developed using PHP, MySQL, HTML, CSS, and session-based authentication. The project includes user management, shopping cart functionality, product management, and an administrative dashboard.

## Project Overview

The application enables users to register, log in, browse products, manage shopping carts, and place orders. Administrators can securely manage products through a dedicated dashboard.

## Key Features

- User Registration & Login
- Secure Password Hashing
- Session Management
- Product Catalog Management
- Shopping Cart System
- Admin Dashboard
- Product Image Uploads
- Database Integration using MySQL and PDO

## Technology Stack

Frontend: HTML, CSS  
Backend: PHP  
Database: MySQL  
Authentication: PHP Sessions & Password Hashing  
Development Environment: XAMPP / Localhost

## Database Design

Users Table: Stores customer and administrator accounts.  
Products Table: Stores product details, descriptions, prices, and images.  
Cart Table: Tracks products added to user carts and quantities.

## Application Modules

1. Registration Module
2. Authentication Module
3. Product Listing Module
4. Shopping Cart Module
5. Admin Login Module
6. Product Management Module
7. Session & Security Module

## Security Measures

Passwords are hashed before storage. Access to protected pages is controlled using session validation. Administrative functions require admin authentication.

## GitHub Portfolio Summary

This project demonstrates full-stack web development concepts including database design, authentication, CRUD operations, session handling, and role-based access control. It highlights practical implementation of e-commerce workflows and secure web application development.